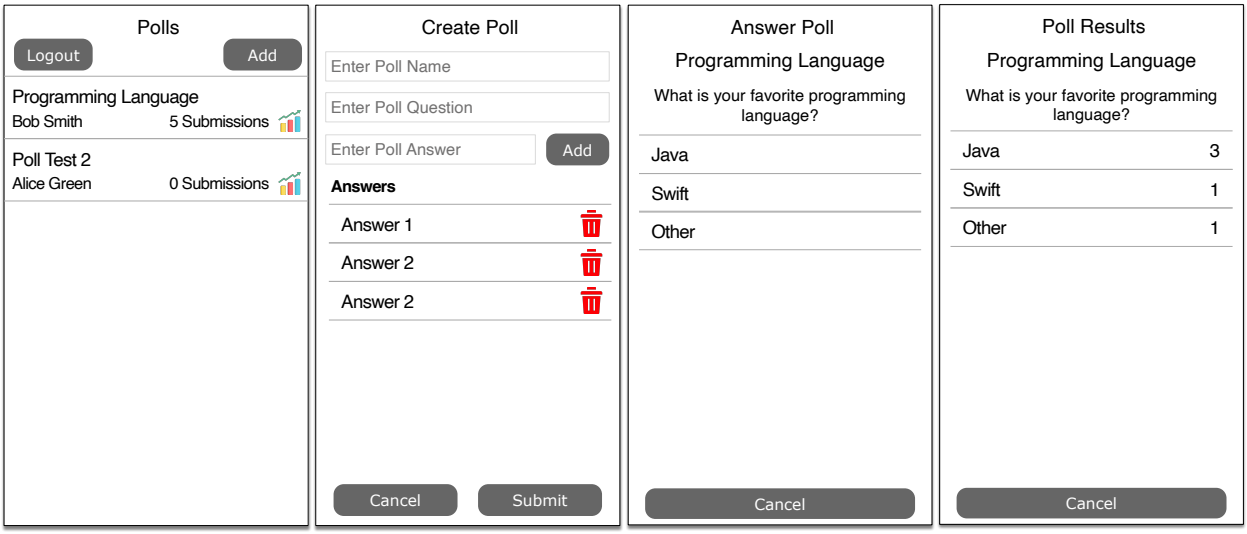
Mobile App Development
In-Class Assessment 3

Basic Instructions:

1. This is and In Class Assessment, which will count for 8% of the total course grade.
2. This assessment is an individual effort. Each student is responsible for her/his own assessment and its submission.
3. Once you have picked up the assessment, you may not discuss it in any way with anyone until the assessment period is over.
4. During the assessment, you are allowed to use the course videos, slides, and your code from previous home works and in class assignments. You can use the internet to search for answers. You are NOT allowed to use code provided by other students or solicit help from other online persons.
5. Answer all the assessment parts, all the parts are required.
6. During the assessment the teaching assistants and Instructors will pass by each student and ask them to demonstrate their application. Your interaction with the teaching assistants and instructors will be taken into consideration when grading your assessment submission.
7. Please download the support files provided with the assessment and use them when implementing your project.
8. Your assignment will be graded for functional requirements and efficiency of your submitted solution. You will loose points if your code is not efficient, does unnecessary processing or blocks the UI thread.
9. Create a zip file which includes all the project folder, any required libraries, and your presentation material. Submit the exported file using the provided canvas submission link.
10. **Do not try to use any Social Messenger apps, Emails, Or Cloud File Storage services in this exam.**
11. **Failure to follow the above instructions will result in point deductions.**
12. **Any violation of the rules regarding consultation with others will not be tolerated and will result disciplinary action and failing the course.**

In-Class Assessment 3 (100 Points)

In this assignment you will be building an application that uses Firebase Firestore Database to manage polls. Each poll has a title, question and a list of questions.



(a) Polls **(b) Create Poll** **(c) Answer Poll** **(d) Poll Results**

Figure 1, Application User Interface

Part 0, Firebase Library Setup (0 Points):

1. We included the required Firebase libraries to the provided project.
2. Add `google-services.json` or `GoogleService-Info.plist` file into project.
3. The login and signup code has been included in your project.

Part 1, Polls Screen (30 Points):

This screen displays the list of polls as shown in Figure 1(a). Requirements are:

1. This screen should retrieve the list of polls from Firestore:
   a. Setup a snapshot listener to listen to realtime updates for the polls collection and to update the list of polls when the polls are updated.
   b. Each row item should display the poll name, owners name, number of competed poll submissions.
2. Clicking the “Add” button should replace the current screen with Create Poll Screen.
3. Clicking a row item should:
   a. If the user has already answered this poll, then display an alert message indicating that the user has already answered this poll.
   b. If the user did not answer this poll, then replace the current screen with the Answer Poll screen also send it the selected poll object.
4. Upon clicking the stats icon:
   a. Replace the current screen with the Poll Results screen also send it the selected poll object.

Part 2, Create Poll Screen (30 Points):

This screen is used to create a new poll. The requirements are listed below:

1. You should be able to enter the poll name and question.
2. You should add at least 2 poll answers:

   a. Entering poll answer and then clicking the Add button should: add the answer to list of answers and then refresh the displayed list below to show the added answers.
   b. Clicking the trash can should delete the answer from the list and refresh the displayed list.
3. Clicking on the Submit button should:
   a. Check that the name, and question are entered. Also make sure the user has added at least 2 answers in order to proceed. If these criteria are met then save the poll and answers on Firebase Firestore.
   b. If any of the entries are not entered or selected then show an alert message indicating the missing input.

Part 3, Answer Poll Screen (30 Points):
This screen allows a user to answer the poll. The requirements are listed below:
1. Each user should be able to answer a poll once and not multiple times.
2. Display the poll name, question and list of answers as shown in Figure 1(c).
3. Clicking on an answer row item:
   a. Send the selected answer to Firebase to record the user's poll answer.
   b. Go back to the Polls screen.

Part 4, Poll Results Screen (10 Points):
This screen displays the poll and the statistics of the selected answers. The requirements are listed below:
1. Display the poll name, question and list of answers as shown in Figure 1(d).
2. Beside each answer display the number of users who have selected this answer.

Section:	
Student Name:	
Student ID:	

Part #	Features	Total	Grade	Comments
Part 2	- Implements the UI and logic - Stores the poll and answers in a document under the polls collection	30		
Part 1	- Setup the SnapShotListener to retrieve the polls and displays the data as shown in Figure 1(a).	20		
Part 3	- Displays the poll information and answers as shown in Figure 1(c). - Clicking on an answer row updates the poll to record the user's answer.	30		
Part 1	- A user is only able to answer a poll once. An alert is displayed if a user attempts to answer a poll they already answered.	10		
Part 4	- Displays the poll information, answers and stats as shown in Figure 1(d).	10		
	Total	**100**		

Table 1: Grading Key